

XR for Mental Health: How Users Perceive New Technologies?

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ABSTRACT

XR technologies can provide effective help to people with mental health needs. We present ideas and preliminary studies on how people understand new XR technologies and how they can shape the future experiences related to their mental health well-being. We also propose a future where users of these technologies could report their mental health related needs and coordinate efforts between psychologists, app providers, and XR technologies in order to fulfill such needs.

Index Terms: XR, Mental Health, Codesign

1 INTRODUCTION

Extended Reality (XR) technologies continue their progress in different directions, such as market share, available options, and overall user experience. One of the fields where XR could be beneficial for humanity is mental health. From the point of view of its common use and the availability of special features for treatment, XR is a technology that should take into account the special needs related to mental health that many people in the world have.

We have carried out research in the field of technology and mental health with different technologies and for different populations [8, 2]. In this work, we emphasize the role that users could have in the development of new functionality in XR environments, and show preliminary results on how existing XR applications could help users with their mental health problems. We also describe how mental health should be an intrinsic feature of these technologies.

2 RELATED WORK

The application of XR technologies for mental health has gained significant traction, with evidence suggesting their utility in therapeutic interventions, education, and patient engagement. We show here advances in XR health applications, focusing on mental health treatments and the unique opportunities presented by new devices such as the Apple Vision Pro (AVP). In addition, we identify gaps where XR can better address psychological well-being.

XR technologies, particularly Virtual Reality (VR) and Augmented Reality (AR), have shown promise in addressing mental health conditions such as anxiety, depression, and PTSD [10]. Riches et al. [12] presented a systematic review of VR relaxation techniques for mental health conditions, highlighting their effectiveness in reducing stress and promoting relaxation. These interventions often involve immersive, nature-based environments like forests, beaches, and mountains to provide calming experiences,

which are shown to be as effective, if not superior, to non-VR relaxation exercises.

In addition to relaxation techniques, XR technologies facilitate novel approaches for mental health treatment such as Cognitive Behavioral Immersion (CBI). Studies by Ezawa et al. [7] and Robinson et al. [13] demonstrate how XR can deliver accessible, lay-coach-led sessions that improve symptoms outcomes, particularly for anxiety and depression. These findings underscore XR's unique ability to engage users actively while delivering structured therapeutic exercises.

Charlotin [4] explores how the Apple Vision Pro can bridge traditional therapy with XR solutions. AVP features guided meditation programs, virtual therapy sessions, and mood tracking functionalities, which collectively enhance emotional regulation through exposure therapy and mindfulness activities. These functionalities highlight the role of XR in augmenting cognitive behavior strategies to improve patient outcomes. Other authors such as Waisberg et al. [14] highlight AVP's ability to deliver immersive clinical simulations for anatomy education and surgical training. Waisberg et al. [15] also emphasize AVP's potential for vision screening, rehabilitation, and precision training in ophthalmology. However, despite its significant utility in medicine, its potential to address mental health needs remains underexplored.

Although XR technologies offer considerable benefits for mental health interventions, it is crucial to incorporate user-centered design principles to maximize their effectiveness. Paquin et al. [9] stress the importance of balancing immersive experiences with safety considerations, particularly in mitigating risks such as escapism or addiction. These findings highlight the need for ethical design processes that prioritize user's well-being.

In this context, co-design methodologies emerge as a promising solution. Engaging end-users as well as patients and psychologists in the development of XR applications ensures that therapeutic tools align closely with the needs of mental health patients. Robinson et al. [13] further emphasize the importance of structured programs to maintain the therapeutic focus of XR interventions, preventing unwanted negative effects.

It is also essential to consider potential drawbacks. Di Pomponio and Cerniglia [6] highlight the dual nature of XR technologies. Although immersive environments can offer therapeutic benefits, excessive reliance on immersive environments can lead to avoidance behaviors or addiction. This underscores the need for ethical and mindful design practices that strike a balance between immersion and user safety.

Despite significant advancements, the application of XR technologies for mental health remains underdeveloped compared to other domains, such as education and productivity. Cheng et al. [5] explore AVP's immersive telepresence features, which aim to improve virtual communication and productivity. Although these features address challenges such as "Zoom fatigue," they do not focus on mental wellness.

Furthermore, although AVP includes mindfulness applications as part of its default system, the broader potential to integrate mental health tools into operating systems and hardware remains unexplored. This gap highlights an opportunity for XR technologies to

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prioritize mental health features, much like accessibility options in current consumer devices.

3 PRELIMINARY STUDIES

We conducted two preliminary studies to better understand how XR technologies can support mental health, and how users could participate in the evaluation and development of XR experiences that could help with mental health issues. The first study gathered opinions from individuals with prior experience with XR about their preferences and perceptions on the use of XR applications for mental health. The second study focused on comparing user experiences between two XR devices, the Quest 3 and the Apple Vision Pro. This work was performed under the supervision of the ethics committee of the University of La Laguna with the report with registration number CEIBA2024-3470.

3.1 Opinions from Users with Previous XR Experience

To evaluate user experiences with XR applications, we conducted a survey aimed at people aged 15 to 35 years with prior XR experience. The survey evaluated various aspects, including mood-lifting potential, perceptions of VR's potential for mental health, usability, and challenges during use. A psychologist and us selected four XR applications with potential for mental health treatment. We show videos of such applications to participants and ask them to provide open feedback on positive and negative aspects of treatment.

3.1.1 Application Preferences and Descriptions

26 participants choose their preferred VR application for mood enhancement from four options. The results are shown in Figure 1, which highlights the distribution of preferences among the participants.

- **The Thrill of the Fight:** This VR boxing game emerged as the most preferred application (42.3%). Known for its authenticity, it places players in a virtual gym where they battle challengers with unique styles that require timing, skill, and strategy to succeed. The participants highlighted that energetic participation and physical activity are key factors in improving mood.
- **TRIPP:** Ranked second (34.6%), this meditative application helps users manage emotional well-being through guided meditation, mood tracking, and personalized VR experiences. The participants valued its soothing effects and its ability to enhance focus.
- **The Room VR:** Favored by 15.4% of the participants, this narrative-driven puzzle game immerses users in an archaeological mystery. The combination of cognitive challenges and immersive storytelling appealed to specific user preferences.
- **Nature Treks:** The least preferred application (7.7%) offers tranquil experiences such as exploring tropical beaches, underwater oceans, and outer space. Although appreciated for relaxation and immersion, it was seen as less engaging compared to other applications.

3.1.2 Positive Aspects of Using VR for Mental Health

Qualitative responses were classified into recurring themes, highlighting the following benefits.

- **Immersion and Multi sensory Experiences:** Eight participants highlighted the potential of virtual reality environments to provide a deep sense of presence and engagement. They noted how VR could facilitate temporary disconnection from real-world stressors.

Application Preferences for Mood Enhancement

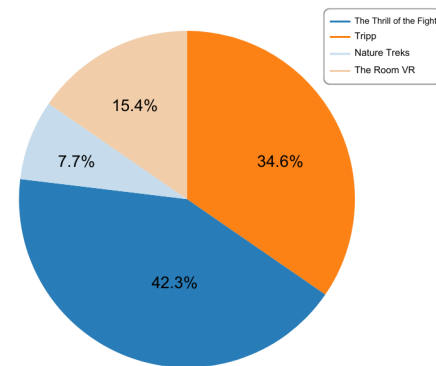


Figure 1: Application Preferences for Mood Enhancement

- **Variety and Versatility:** Four participants appreciated the diversity of applications, emphasizing that VR can support recreation, meditation, and exercise, making it adaptable to a range of therapeutic needs.
- **Stress Reduction and Therapeutic Benefits:** Thirteen participants reported that the applications conveyed a strong capacity to reduce anxiety and improve mood, reflecting the direct mental health benefits of VR.
- **Innovation and Improvement Potential:** One participant highlighted VR's novelty as a scalable mental health tool, with the potential for widespread institutional and personal applications, as shown in the examples.

3.1.3 Negative Aspects of Using VR for Mental Health

The participants also noted several challenges and limitations, such as the following:

- **Physical Discomfort:** Eight participants expressed concerns about potential issues such as dizziness, eye strain, and mild nausea, which were inferred as possible drawbacks of prolonged VR use.
- **Dependency and Disconnection:** Eight responses raised concerns about the potential for excessive use, dependency, and avoidance of real-world problems, emphasizing the need for responsible use.
- **Therapeutic Focus:** Four participants suggested that some applications could deviate from therapeutic objectives, highlighting the importance of designing focused and goal-oriented VR experiences.
- **Accessibility Barriers:** Two participants identified high costs and limited access as significant obstacles to the adoption of VR for mental health, underscoring the need for affordability and inclusion.

3.1.4 Demographics and Mood

The median age of the participants was 22 years, w. On a mood scale (1 = depressed, 7 = happy), 52% of the participants scored 5, indicating moderate happiness. Participants reported moderate experience with VR technologies (median score: 5) and video games (median score: 6). Both expertise scales ranged from 1 (minimal experience) to 7 (expert level).

3.1.5 Discussion

The results emphasize VR's versatility as a tool for mental health, with *The Thrill of the Fight* excelling in active stress relief and *TRIPP* providing a calming alternative. These applications cater to diverse therapeutic needs, from physical engagement to relaxation and emotional regulation.

However, challenges such as physical discomfort and concerns about dependency underscore the need for thoughtful design. Tailoring applications to user preferences and addressing accessibility barriers are critical for maximizing VR's therapeutic potential. Despite the demographic limitations of this study, the high receptivity to VR indicates a promising avenue for mental health interventions.

Although there are several approaches in the literature for some of the negative aspects found, we plan to run more thorough studies in order to identify the most important concerns, in order to prioritize our future research work. In particular, solutions for technological dependency has been studied [3] in the past, and these results could be applied to these new technologies.

3.2 Comparison by Novices

This study involved a group of six volunteer students aged between 22 and 25 years old. The primary objectives of this study were to:

- Identify differences and preferences in user experiences with two XR devices, the Quest 3 and the AVP.
- Collect participant opinions on the potential for developing a depression treatment application using these technologies.
- Explore the feasibility of extending the study into a more prolonged and generalized co-design task.

3.2.1 Procedure

Before the experimental session, participants attended a 10-minute group meeting where the study objectives were outlined, and informed consent was obtained. During this session, the Technology Acceptance Model (TAM) questionnaire and the Beck Depression Inventory of 21 items [1] were administered to assess initial attitudes toward technology and identify depression symptoms among participants.

Afterwards, in pairs, participants participated in a scheduled 40 minutes demonstration session. During these sessions, participants were introduced to the Quest 3 and the AVP in a counterbalanced order to minimize device-order bias. Depending on the device, participants engaged with a relaxation application, such as *Tripp* or *Mindfulness*. If time allowed, participants were encouraged to explore additional applications, such as *Nature Treks*, the AVP environments, or XR games, to assess further the devices' capabilities and usability.

Following the demonstration sessions, participants performed a post-test evaluation. They completed the Igroup Presence Questionnaire (IPQ) [11] to assess their immersion and presence within the virtual environments. Subsequently, a post-test questionnaire was administered, which included the TAM to measure changes in technology acceptance. This questionnaire also featured exploratory questions (see Annex) aimed at evaluating the potential of VR for depression treatment and gathering input for future co-design initiatives.

3.2.2 IPQ Results

The radar chart presented in Figure 2 compares the results of the AVP and Quest glasses in three main dimensions of the IPQ Questionnaire: Involvement (INV), Spatial Presence (SP), and Experienced Realism (REAL). These results provide a detailed view of users' perceptions regarding the effectiveness of each device in creating an immersive and realistic experience.

Comparison between AVP and Quest IPQ results

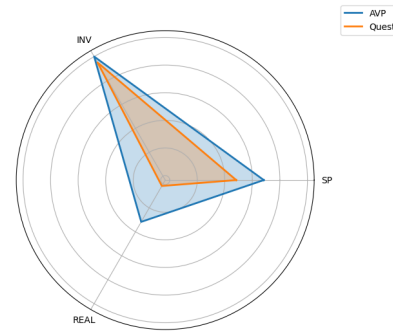


Figure 2: Comparison between AVP and Quest IPQ Results

The AVP had the highest score on the Experienced Realism (REAL) axis, indicating that users perceived more realism using this device. The blue line, representing AVP, reaches the maximum value on this axis, suggesting that users felt significantly more realistic about the virtual environment provided by AVP. In comparison, the Quest glasses, represented by the orange line, scored lower on the same axis, indicating a lower level of realism.

The AVP continued to outperform the Quest glasses for the Spatial Presence (SP) axis, though with a less pronounced difference than the previous axis. The intermediate score on this axis shows that while AVP provided a convincing sense of spatial presence, the Quest glasses were also effective but slightly disadvantaged. This result suggests that both devices are competent in creating spatial presence, but AVP still offers a marginal advantage.

Both lines showed a more balanced performance on the Involvement (INV) axis, with AVP still leading but with a smaller margin. This result suggests that both devices effectively create a realistic virtual experience.

In conclusion, the Apple Vision Pro demonstrated superior performance in all three dimensions. The results indicate that AVP provides a more engaging and realistic virtual reality experience than Quest glasses.

Despite the superior results of the Apple Vision Pro, it must be recognized that both devices seem to be effective in providing an immersive and realistic experience. The slight superiority of AVP may be associated with factors such as advanced display technology, integration of more precise sensors, novelty, or a more intuitive user interface. Nevertheless, it is important to deepen future investigations to gain a more complete and detailed understanding of performance differences. Additional studies may consider application diversity, exposure time, and individual user preferences. Thus, it will be possible to further refine both technologies and expand their therapeutic potential in virtual reality.

3.2.3 Answers to the Exploratory Questions

Overall, the feedback obtained by the exploratory questions was predominantly positive, with participants consistently highlighting the immersive nature of the VR experience. This immersion facilitated a sense of presence within the virtual environments, often leading to feelings of relaxation and tranquility post-session. Participants generally found the VR equipment easy to use, particularly the AVP, with only minor challenges reported during the initial familiarization with certain interactive applications. The potential of VR as a therapeutic tool for depression was widely acknowledged, primarily due to its capacity to provide distraction, promote relaxation, and offer a sense of control within a controlled virtual setting.

Participants mentioned the following positive aspects:

- **Immersion and Presence:** A recurring theme in the responses of the participants was the strong sense of immersion and presence experienced within VR environments. 360-degree videos were particularly effective in creating a feeling of 'being there', with some participants reporting a temporary detachment from the real world. This immersive quality is crucial for therapeutic applications, as it allows people to engage with virtual experiences in a meaningful way.
- **Relaxation and Mood Improvement:** The participants reported a noticeable improvement in their mood after VR sessions, primarily characterized by feelings of relaxation and reduced anxiety. This suggests that VR can effectively promote relaxation and potentially mitigate stress and low-mood symptoms.
- **Ease of Use:** The participants found the VR equipment user-friendly, especially the AVP. This ease of use is essential for the widespread adoption of VR technology in therapeutic settings, as it reduces barriers to entry for individuals unfamiliar with the technology.
- **Therapeutic Potential for Depression:** Participants expressed a strong belief in the potential of VR as a complementary therapy for depression. They highlighted the ability of VR to provide a safe and controlled environment for exploring emotions, practicing coping mechanisms, and experiencing positive stimuli.

Participants also mentioned the following issues for improvement:

- **Navigation and Movement:** One participant reported difficulties with navigation and movement within the virtual environment, suggesting that further refinement of the VR interfaces and control mechanisms may be required to ensure a seamless and comfortable user experience.
- **Auditory Immersion:** Several participants suggested enhancing the auditory aspect of the VR experience by incorporating noise canceling headphones, isolating users from external distractions and deepening their sense of immersion. The capability of controlling ambient sound within the VR experience was also suggested.
- **Motion Sickness/Vertigo:** One participant experienced a mild sensation of vertigo during a VR session, particularly while standing, highlighting the importance of considering individual sensitivities and providing options for users to engage with VR in a seated position or with applications based on less intense movement.
- **Need for Tailored Experiences:** One participant pointed out the need for more customized experiences, suggesting that asking specifically about different types of VR experience (e.g., games, guided meditation, exploration) would provide more nuanced feedback. This underscores the importance of developing diverse VR applications to cater to individual needs and preferences.

4 ENVISIONING THE FUTURE

We think mental health should be protagonist in any new technology, in particular XR. This is also evident to some companies. For example, the Apple Vision Pro includes a default application for mindfulness, showing the importance of the users' well-being at any time. Mental health should be a priority for XR technologies, much like accessibility is currently for most of the devices that a

consumer can buy. Actions can be taken by all actors involved, as follows.

Producers of XR hardware should include in their operating systems (OS) configurations and features related to mental health. Much like we have now accessibility options at the level of a system, XR hardware should collect information and provide assistance related to mental health in a private way. There are already examples of such functionality. For example, the Health application in iPhone or iPad allows users to register their mood at any time, see summaries of such information, and share such information with relatives. Such functionality should be kept private but also accessible at will to authorized parties, and therefore such functionality should be integrated at the level of the operating system, not at the mercy and duplication of one or several applications.

There also opportunities for XR Applications in mental health. XR applications should take into account the generic information that a device could collect in relation with mental health, and also help to collect such information. For example, behavioral changes detected in an application could be registered in the general mental health service of the operating system, and in this way help other applications and services to accommodate to the user's state of mind.

XR users play also an active role in the development of XR for mental health. Questionnaires such as the ones used in our methodology could be part of the user experience of most XR applications, if not all. They could also be integrated to the OS services in mental health. In this way, XR developers could be informed of the mental health implications of their applications, and include new functionality related to these issues once it is requested by users.

Psychologists in general could also be informed of the potential of XR technologies for treatment of mental health conditions. There are important research results in new topics like cyberpsychology and cybertherapy, and more should be done in these directions. More education and more availability could help in order to make XR technologies mainstream.

Finally, Advocacy groups could be created in order to advance the support for mental health in XR. Research related groups could lead the way, but more diverse groups are needed in order to suggest future policies in different contexts.

5 CONCLUSIONS AND FUTURE WORK

We envision a future where XR technologies support real treatments for mental health conditions, and promote a healthy state of mind in their users. Users could be an active part of such future, by informing developers their needs and preferences related to mental health. More research is necessary to assess this possibility. Finally, coordination between different actors and the use of techniques such as co-design could help in the development of real solutions and ecosystems.

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REFERENCES

- [1] A. T. Beck, C. H. Ward, M. Mendelson, J. Mock, and J. Erbaugh. An inventory for measuring depression. *Archives of general psychiatry*, 4(6):561–571, 1961. 3
- [2] A. F. Bejarano, J. D. Correa, and P. Figueroa. Escape room virtual reality: A tool for diagnosis and treatment of attention deficit disorder. In *2020 22nd Symposium on Virtual and Augmented Reality (SVR)*, pp. 331–338, 2020. doi: 10.1109/SVR51698.2020.00056 1
- [3] P. Berthon, L. Pitt, and C. Campbell. Addictive de-vices: A public policy analysis of sources and solutions to digital addiction. *Journal*

- of Public Policy & Marketing, 38(4):451–468, 2019. doi: 10.1177/0743915619859852 3
- [4] C. Charlotin. Apple vision pro therapy tool: A new frontier in mental health therapy. <https://www.drcpsychologist.com/post/apple-vision-pro-therapy-tool-mental-health>, 2024. 1
 - [5] R. Cheng, N. Wu, M. Varvello, E. Chai, S. Chen, and B. Han. A first look at immersive telepresence on apple vision pro. In *Proceedings of the 2024 ACM on Internet Measurement Conference*, pp. 555–562, Nov. 2024. doi: 10.1145/3584825.3601252 1
 - [6] I. Di Pomponio and L. Cerniglia. Exploring the mental health frontier: Social media, the metaverse and their impact on psychological well-being. *Adolescents*, 4(2):226–230, 2024. doi: 10.3390/adolescents4020016 1
 - [7] I. D. Ezawa, S. D. Hollon, and N. Robinson. Examining predictors of depression and anxiety symptom change in cognitive behavioral immersion: Observational study. *JMIR Ment Health*, 10:e42377, 2023. doi: 10.2196/42377 1
 - [8] P. Figueroa, C. Rivera, P. Casas, J. López, D. Mora, I. Gomez, and K. Peñaranda. A case study on user centered design for vr based training in health. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1066–1067, 2024. doi: 10.1109/VRW62533.2024.00329 1
 - [9] V. Paquin, M. Ferrari, H. Sekhon, and S. Rej. Time to think 'meta': a critical viewpoint on the risks and benefits of virtual worlds for mental health. *JMIR Serious Games*, 11:e43388, 2023. doi: 10.2196/43388 1
 - [10] P. Pons, S. Navas-Medrano, and J. L. Soler-Dominguez. Extended reality for mental health: Current trends and future challenges. *Frontiers in Computer Science*, 4, 2022. doi: 10.3389/fcomp.2022.1034307 1
 - [11] H. Regenbrecht and T. Schubert. Real and illusory interactions enhance presence in virtual environments. *Presence: Teleoperators & Virtual Environments*, 11(4):425–434, 2002. 3
 - [12] S. Riches, P. Jeyarajaguru, L. Taylor, C. Fialho, J. Little, L. Ahmed, and L. Valmaggia. Virtual reality relaxation for people with mental health conditions: a systematic review. *Social Psychiatry and Psychiatric Epidemiology*, 58(7):989–1007, 2023. doi: 10.1007/s00127-022-02417-5 1
 - [13] N. Robinson, A. Mahapatra, B. Jean-Baptiste, A. Mallard, A. Yang, S. D. Hollon, and I. D. Ezawa. Cognitive behavioral immersion for substance use disorders: A feasibility and pilot study of a peer-based coaching program in the metaverse. *Games for Health Journal*, 12(5):397–404, 2023. doi: 10.1089/g4h.2022.0214 1
 - [14] E. Waisberg, J. Ong, M. Masalkhi, N. Zaman, P. Sarker, A. G. Lee, and A. Tavakkoli. Apple vision pro and the advancement of medical education with extended reality. *Canadian Medical Education Journal*, 15(1):89, 2024. doi: 10.36834/cmej.74221 1
 - [15] E. Waisberg, J. Ong, M. Masalkhi, N. Zaman, P. Sarker, A. G. Lee, and A. Tavakkoli. The future of ophthalmology and vision science with the apple vision pro. *Eye*, 38(2):242–243, 2024. doi: 10.1038/s41433-023-02480-9 1
 7. Do you have any suggestions for improving the VR experience?
 8. How did your expectations about VR change after participating in the experiment?
 9. Would you recommend the use of VR to other young university students dealing with depression?
 10. How would you describe the overall impact of the VR session on your well-being?
 11. Is there anything else you would like to add about your experience with VR and its potential use in treating depression?

ANNEX - EXPLORATORY QUESTIONS

1. Describe your experience using the VR headset. What did you like the most, and what didn't you like?
2. How would you describe the sensation of being in the virtual environment? Did you feel present and immersed?
3. Did you notice any changes in your mood after the VR session? Please describe.
4. Do you think VR can be an effective tool for treating depression? Why?
5. How would you evaluate the ease of use of the VR headset? Did you experience any difficulties?
6. Was there anything about using the VR headset that caused you discomfort?